

JET-GRAM STRUCTURES, CANONICAL SUBSPACE INVARIANTS, AND HIGHER QUANTUM GEOMETRY

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ABSTRACT. The quantum geometric tensor is the first-order Gram matrix of horizontal derivatives of a pure-state family. Motivated by the local jet cross-Gram pattern in the Riemann-side construction, we ask whether this first-order geometric object has a higher-order, two-point matrix analogue for quantum state families. The answer is sharply mixed. At first order the same-point horizontal Gram matrix is exactly the usual quantum geometric tensor, and a transported two-point first-jet block has a precise reparameterization law. From second order onward, however, ordered jet matrices transform by non-unitary triangular laws under phase gauge and coordinate change. Their entries are not canonical. The surviving objects are the horizontal osculating subspaces O_r , together with the ambient extensions $A_r = \text{span}\{\psi\} \oplus O_r$. Endpoint comparison is then governed by cross-contractions, projector compressions, polar partial isometries, and principal angles. We prove that all orthonormal-frame comparison matrices for a fixed endpoint subspace pair form one biunitary orbit, classified by the same principal-angle singular values; a canonical entrywise matrix therefore requires extra frame-selecting structure. We give polynomial and non-polynomial benchmarks where the resulting higher-order subspace data is richer than ordinary overlap, record a finite-jet genericity statement for the first horizontal line invariant, and describe a coherent reflection protocol that reads the principal angles under explicit access assumptions. Research gaps are isolated where stronger physics-facing claims would require new input.

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1. INTRODUCTION

The usual quantum geometric tensor is a first-order object. Given a smooth family of normalized pure states, one removes the component of a derivative in the state direction and forms the Gram matrix of the resulting horizontal derivatives. The real part is the quantum metric; the imaginary part is Berry curvature. This paper asks what survives when the same Gramian idea is pushed to higher jets and to two endpoint comparisons.

The motivating algebraic pattern is a local jet cross-Gram block. One has same-point Gram matrices at two nearby points, a mixed cross block between the two local jet frames, and a bilateral whitening. In abstract notation this has the form

$$\Omega = M_-^{-1/2} N_{-+} M_+^{-1/2}.$$

The question is not whether a particular kernel-theoretic object is literally a standard quantum tensor. The question is narrower and more structural:

Which parts of a higher-jet Gram construction are canonical
for quantum state families, and which parts depend on phase,
coordinates, frames, or transport?

The answer is the main message of the paper. At first order one obtains the standard quantum geometric tensor and a controlled two-point transported block. At higher order the matrix entries do not survive. The ordered

derivative list mixes by upper-triangular, non-unitary transformations under ordinary changes of phase and parameter. Whitening does not repair this. The canonical survivor is the subspace spanned by the derivatives. Comparing endpoints means comparing subspaces.

Thus the central object is not a higher quantum geometric matrix, but the endpoint operator package of two osculating subspaces:

$$C = \Pi_-|_{S_+}, \quad K_+ = C^*C, \quad K_- = CC^*, \quad C = V|C|.$$

Its spectra give the squared cosines of principal angles. The polar factor gives a canonical partial isometry on the nonzero-overlap sector. The principal-angle multiset classifies finite-dimensional subspace pairs up to ambient unitary equivalence.

This package has two uses. First, it gives a higher-order geometric fingerprint of a state family. In explicit qutrit, quartit, Veronese, and spherical-twist families it distinguishes endpoint pairs with the same ordinary overlap. Second, it gives a no-go criterion. A proposed higher-jet comparison matrix is not canonical unless the extra structure selecting its frames is stated.

1.1. Main contributions. The paper proves or records the following claims.

- (1) The same-point first horizontal Gram matrix is exactly the pure-state quantum geometric tensor.
- (2) A one-parameter first-horizontal two-point block is invariant under orientation-preserving reparameterization after a fixed gauge-covariant transport choice, and transforms by derivative-channel sign conjugation under orientation reversal.
- (3) Raw higher projected jets transform by upper-triangular laws under phase gauge and reparameterization. Therefore their Gram and whitened matrices are not canonical.
- (4) The horizontal osculating spaces O_r and the ambient spaces A_r are canonical subspace-valued invariants.
- (5) Berry-covariant one- and multi-parameter jet hierarchies generate the same O_r filtrations as raw projected jets. Symmetrized covariant jets generate the same multiparameter filtration.
- (6) Endpoint comparison is controlled by cross-contractions, projector compressions, polar partial isometries, and principal angles.
- (7) Every orthonormal-frame comparison matrix for a fixed subspace pair lies in a single biunitary orbit; the rectangular singular values are exactly the principal-angle cosines.
- (8) Canonical principal frames exist from the subspace pair alone exactly on simple nonzero principal spectra. Repeated principal values have an unavoidable internal unitary freedom.
- (9) Exact unitary orbits identify the ambient and horizontal spaces with transported Krylov and compressed-Krylov spaces. Kato transport

gives a pathwise realization but not endpoint-independent transport in nonzero curvature.

- (10) With coherent reflection access to endpoint projectors, phase estimation on $(2\Pi_- - I)(2\Pi_+ - I)$ reads the same principal angles.
- (11) Benchmark families show richer-than-overlap behavior. A finite-jet genericity criterion gives a first-order mechanism behind this behavior.

1.2. What is not claimed. The paper does not claim the discovery of new fundamental particles. It does not claim that higher jets define a canonical matrix. It does not claim that the construction is fundamental for all quantum mechanics. It proves a mathematical invariant package and marks the additional work needed to turn that package into a physics-facing diagnostic in concrete systems.

2. BACKGROUND AND NOTATION

Let $\mathcal{H}$ be a complex Hilbert space. Most statements below are cleanest in finite dimension; the subspace and projector statements extend to closed finite-rank subspaces in a Hilbert space with the usual domain qualifications. The inner product is linear in the second variable only if a convention is chosen; the formulas below are invariant after conjugating consistently. We write $|\psi\rangle$ only in informal passages and use ψ in formulas.

A smooth pure-state family is a smooth map $x \mapsto \psi(x) \in \mathcal{H}$ with $\|\psi(x)\| = 1$. The physical state is the ray $[\psi(x)]$, so the local lift is defined only up to a phase $e^{i\chi(x)}$. The orthogonal projector onto the horizontal space at ψ is

$$P_\psi = I - |\psi\rangle\langle\psi|.$$

For a coordinate vector ∂_a on parameter space, define the horizontal derivative

$$D_a\psi = P_\psi\partial_a\psi.$$

Definition 2.1 (Quantum geometric tensor). *The pure-state quantum geometric tensor of the family $x \mapsto \psi(x)$ is*

$$Q_{ab}(x) = \langle D_a\psi(x), D_b\psi(x) \rangle = \langle \partial_a\psi, (I - |\psi\rangle\langle\psi|)\partial_b\psi \rangle.$$

Its real part is the Fubini-Study metric tensor and twice its imaginary part is the Berry curvature in the convention used here:

$$g_{ab} = \Re Q_{ab}, \quad F_{ab} = 2\Im Q_{ab}.$$

We also need the elementary geometry of two subspaces. If S_- and S_+ are finite-dimensional subspaces with orthogonal projectors Π_- and Π_+ , the canonical cross-contraction is

$$C = \Pi_-|_{S_+} : S_+ \rightarrow S_-.$$

The positive compressions are

$$K_+ = C^*C = \Pi_+\Pi_-\Pi_+|_{S_+}, \quad K_- = CC^* = \Pi_-\Pi_+\Pi_-|_{S_-}.$$

The singular values of C are the cosines of the principal angles between S_- and S_+ .

3. FIRST-ORDER QUANTUM JET-GRAM GEOMETRY

3.1. The same-point bridge. The first-order same-point part of the construction is exactly standard quantum geometry.

Theorem 3.1 (First-order QGT bridge). *Let $x \mapsto \psi(x)$ be a smooth normalized pure-state family. The same-point horizontal derivative Gram matrix*

$$(\langle P_\psi \partial_a \psi, P_\psi \partial_b \psi \rangle)_{a,b}$$

is the pure-state quantum geometric tensor.

Proof. The statement is the definition after horizontal projection. Since $\|\psi\|^2 = 1$, the derivative $\partial_a \psi$ decomposes into the state line and its orthogonal complement. Removing the state-line component gives

$$D_a \psi = (I - |\psi\rangle\langle\psi|) \partial_a \psi.$$

Taking the Gram matrix of these horizontal derivatives gives

$$\langle D_a \psi, D_b \psi \rangle = \langle \partial_a \psi, (I - |\psi\rangle\langle\psi|) \partial_b \psi \rangle,$$

which is the usual expression for the pure-state quantum geometric tensor. $\square$

This theorem is the exact first-order contact with established physics. It does not identify the full two-point whitened block with the quantum geometric tensor; it identifies the same-point horizontal block.

3.2. A transported first-order two-point block. Let $t \mapsto \psi(t)$ be a one-parameter normalized lift. Define

$$h_t = P_t \dot{\psi}(t), \quad P_t = I - |\psi(t)\rangle\langle\psi(t)|.$$

Choose a two-point transport rule $U_{-\leftarrow+}$ from the endpoint fiber at t_+ to the endpoint fiber at t_- . In the ordered pair (ψ, h) , define

$$M(t_i) = \begin{pmatrix} 1 & 0 \\ 0 & \langle h_{t_i}, h_{t_i} \rangle \end{pmatrix}$$

and

$$N(t_-, t_+) = \begin{pmatrix} \langle \psi_-, U_{-\leftarrow+} \psi_+ \rangle & \langle \psi_-, U_{-\leftarrow+} h_+ \rangle \\ \langle h_-, U_{-\leftarrow+} \psi_+ \rangle & \langle h_-, U_{-\leftarrow+} h_+ \rangle \end{pmatrix}.$$

The whitened first-order block is

$$\Omega_t = M(t_-)^{-1/2} N(t_-, t_+) M(t_+)^{-1/2}.$$

Proposition 3.2 (First-order reparameterization law). *Let $\mu = \phi(t)$ be a smooth reparameterization with nonzero derivative at the endpoints. For a fixed transported endpoint comparison as above,*

$$\Omega_\mu = \Omega_t$$

when ϕ is orientation-preserving at both endpoints. If the orientation is reversed at both endpoints and the ordered endpoint pair is kept fixed, then

$$\Omega_\mu = E\Omega_t E, \quad E = \text{diag}(1, -1).$$

Proof. Under reparameterization,

$$h_\mu = P_\psi \frac{\partial \psi}{\partial \mu} = \frac{dt}{d\mu} h_t.$$

The derivative channel at each endpoint is therefore multiplied by the real scalar $dt/d\mu$. The corresponding same-point derivative norm in $M(t_i)$ is multiplied by its square, while the mixed entries involving one derivative channel are multiplied by the endpoint derivative factor. Bilateral whitening cancels the positive magnitudes. Only the signs remain. If both signs are positive, nothing changes. If both endpoint orientations reverse, the derivative channel is conjugated by the diagonal sign matrix $E = \text{diag}(1, -1)$. $\square$

Remark 3.3 (Transport is part of the data). *This proposition is not an endpoint-only theorem. The transport rule is part of the input. If the transport is gauge-covariant, the construction descends from a fixed lift to the corresponding ray curve. In the restricted scalar-phase class $U_{-\leftarrow+} = e^{i\alpha}I$, the whole block changes by a global phase and its singular values are transport-class invariant. Outside such restricted classes, no transport-independent matrix is claimed.*

Research gap 3.4 (GAP-3A: first-order transport classification). *The current draft uses the fixed-transport and scalar-phase statements because they are enough for the main message. A stronger version would classify all transport choices under which the first-order singular-value package is invariant. This is not needed for the paper's core theorem spine.*

4. HIGHER JETS: MATRIX FAILURE AND SUBSPACE SURVIVAL

The first-order matrix behavior is exceptional. Higher derivatives mix under ordinary changes of phase and parameter.

4.1. Triangular transformation laws. For a one-parameter curve define raw projected jets

$$j_k(t) = P_t \partial_t^k \psi(t), \quad 1 \leq k \leq r.$$

Lemma 4.1 (Phase triangularity). *Under a phase change $\psi(t) \mapsto e^{i\chi(t)}\psi(t)$, the ordered list $(j_1, \dots, j_r)$ is transformed by multiplication by $e^{i\chi(t)}$ and an invertible upper-triangular linear combination of the lower jets. In particular,*

$$j_2 \mapsto e^{i\chi}(j_2 + 2i\chi' j_1).$$

Proof. Differentiate $e^{i\chi}\psi$ repeatedly. The k -th derivative is

$$\partial_t^k(e^{i\chi}\psi) = e^{i\chi}\partial_t^k\psi + \sum_{\ell < k} b_{k\ell}(\chi', \dots, \chi^{(k-\ell)})e^{i\chi}\partial_t^\ell\psi + b_{k0}e^{i\chi}\psi.$$

The final state-line term is killed by the horizontal projector. The coefficient of j_k is $e^{i\chi}$, so the transformation is triangular with invertible diagonal. The displayed $k = 2$ formula follows by direct differentiation and projection. $\square$

Lemma 4.2 (Coordinate triangularity). *Let $\mu = \phi(t)$ be a one-parameter coordinate change with $dt/d\mu \neq 0$. Then the ordered list $(j_1, \dots, j_r)$ transforms by an invertible upper-triangular combination. In particular,*

$$j_2^{(\mu)} = \left(\frac{dt}{d\mu}\right)^2 j_2^{(t)} + \frac{d^2 t}{d\mu^2} j_1^{(t)}.$$

Proof. This is the chain rule. The k -th μ -derivative contains $(dt/d\mu)^k \partial_t^k \psi$ plus lower t -derivatives. Projecting horizontally kills only the state-line term, not the lower horizontal jets. The diagonal coefficients are nonzero, so the transformation is invertible and triangular. $\square$

Theorem 4.3 (Higher-jet matrix no-go). *For $r \geq 2$, the raw higher-jet Gram matrices and their bilaterally whitened two-point matrix analogues are not canonical under phase gauge and reparameterization. Their entries depend on the ordered jet representatives.*

Proof. The transformation laws above are generally non-unitary triangular changes of ordered jet coordinates. A Gram matrix transforms by congruence under such a change, and bilateral whitening only normalizes the end-point same-point Gram matrices. It does not make the mixed block invariant under arbitrary non-unitary triangular coordinate changes. Hence the entries of the ordered matrices depend on gauge and coordinate choices. $\square$

4.2. Subspace survival.

Theorem 4.4 (Osculating subspace survival). *For a one-parameter ray curve, the subspace*

$$O_r(t) = \text{span}\{j_1(t), \dots, j_r(t)\}$$

is invariant under local phase gauge and under reparameterization with nonzero derivative. In a fixed ambient Hilbert space the ambient space

$$A_r(t) = \text{span}\{\psi(t), \partial_t \psi(t), \dots, \partial_t^r \psi(t)\} = \text{span}\{\psi(t)\} \oplus O_r(t)$$

is also invariant.

Proof. The ordered jets transform by an invertible triangular matrix. Such a matrix changes the generating list but not its span. Adding the state line gives the ambient space A_r . The equality $A_r = \text{span}\{\psi\} \oplus O_r$ follows because $\partial_t^k \psi = j_k + \langle \psi, \partial_t^k \psi \rangle \psi$. $\square$

This theorem is the pivot of the paper. The failed matrix is replaced by a canonical subspace.

4.3. Covariant jets. The subspace theorem can be generated by covariant derivatives rather than raw projected coordinate derivatives. In one parameter define

$$a(t) = \langle \psi(t), \dot{\psi}(t) \rangle, \quad \nabla_t \xi = P_t(\dot{\xi} - a(t)\xi).$$

Set

$$h_{[1]} = \nabla_t \psi, \quad h_{[k+1]} = \nabla_t h_{[k]}.$$

Proposition 4.5 (One-parameter covariant hierarchy). *The vectors $h_{[k]}$ are gauge-covariant, and*

$$\text{span}\{h_{[1]}, \dots, h_{[r]}\} = O_r(t).$$

Proof. Under $\psi \mapsto e^{i\chi}\psi$, a horizontal vector $\xi \mapsto e^{i\chi}\xi$ is sent by ∇_t to $e^{i\chi}\nabla_t \xi$. Thus the hierarchy is gauge-covariant. Expanding $\nabla_t h_{[k]}$ shows that $h_{[k+1]}$ equals j_{k+1} plus a linear combination of lower j_ℓ 's. The change of generators is again triangular with nonzero diagonal, so the spans agree. $\square$

In several parameters, write

$$a_a = \langle \psi, \partial_a \psi \rangle, \quad \nabla_a \xi = P(\partial_a \xi - a_a \xi).$$

Ordered iterated covariant derivatives $\nabla_{a_k} \cdots \nabla_{a_1} \psi$ generate the same O_r spaces as raw mixed projected partial derivatives of total order at most r . Since commutators of the ∇_a 's contribute only lower-order terms on the horizontal bundle, symmetrized covariant derivatives generate the same filtration. Thus the multiparameter hierarchy determines filtered subspaces, not a canonical ordered frame.

Research gap 4.6 (GAP-4A: associated-graded symbols). *The multiparameter no-go leaves room for a basis-free associated-graded or tensor-symbol package. Such an object would not be an endpoint comparison matrix, but it might be a useful local invariant. A future version should either define this object and prove its transformation law, or explicitly leave it out of scope.*

5. ENDPOINT OPERATOR PACKAGE

Once the canonical objects are endpoint subspaces, the comparison problem is classical and rigid.

5.1. Canonical contractions.

Theorem 5.1 (Endpoint operator package). *Let S_- and S_+ be finite-dimensional subspaces of $\mathcal{H}$, with orthogonal projectors Π_- and Π_+ . The following data are canonical:*

$$C = \Pi_-|_{S_+} : S_+ \rightarrow S_-,$$

$$K_+ = C^*C = \Pi_+\Pi_-\Pi_+|_{S_+}, \quad K_- = CC^* = \Pi_-\Pi_+\Pi_-|_{S_-},$$

and the polar decomposition

$$C = V|C|$$

on the nonzero-overlap sector. The nonzero spectra of K_+ and K_- are the squared cosines of the principal angles between S_- and S_+ .

Proof. The projectors are determined by the subspaces, so C , K_+ , and K_- are determined by the subspace pair. The nonzero singular values of C are the square roots of the nonzero eigenvalues of C^*C and CC^* . By the standard definition of principal angles, these singular values are $\cos \theta_i$. The polar decomposition of a bounded finite-rank operator is unique on $(\text{Ker } C)^\perp$, so the partial isometry V is canonical on the nonzero-overlap sector. $\square$

Corollary 5.2 (Projector test). *For a state prepared in S_- , the operator*

$$\Pi_- \Pi_+ \Pi_-|_{S_-}$$

is the effect operator for the sequential yes/yes subspace-membership test. Its eigenvalues are $\cos^2 \theta_i$.

5.2. Frame matrices are orbit representatives. The operator package is often written as a matrix after choosing endpoint frames. The next theorem says exactly what that matrix means.

Theorem 5.3 (Biunitary matrix finality). *Let $U_- : \mathbb{C}^p \rightarrow S_-$ and $U_+ : \mathbb{C}^q \rightarrow S_+$ be orthonormal frame isometries. The comparison matrix*

$$A = U_-^* U_+$$

is the matrix of $C = \Pi_-|_{S_+}$ in those frames. Replacing the frames by U_-L and U_+R , with $L \in U(p)$ and $R \in U(q)$, changes the matrix by

$$A \mapsto L^* A R.$$

Conversely, every orthonormal-frame comparison matrix for the same pair arises in this way. Therefore the singular values of A , equivalently the principal-angle cosines, are the complete frame-independent content of such a matrix.

Proof. Since U_- is an isometry onto S_- , $U_- U_-^* = \Pi_-$ on the ambient space. For $z \in \mathbb{C}^q$,

$$U_-^* C(U_+ z) = U_-^* \Pi_- U_+ z = U_-^* U_+ z,$$

so A is the frame matrix of C . If the frames are replaced by U_-L and U_+R , the new matrix is

$$(U_-L)^*(U_+R) = L^* U_-^* U_+ R = L^* A R.$$

All orthonormal frames of a fixed finite-dimensional subspace differ by a unitary on the coordinate space, so the converse follows. The rectangular SVD classifies the biunitary orbit by singular values, including rank-deficient zeros and rectangular padding. $\square$

Corollary 5.4 (No canonical entrywise matrix without frames). *From a fixed subspace pair alone, a sorted diagonal SVD representative is an orbit normal form, not an endpoint-selected matrix. Any entrywise matrix requires an additional frame-selection rule.*

5.3. When principal frames exist.

Theorem 5.5 (Simple-spectrum frame criterion). *On the nonzero-overlap sector, canonical matched principal frames can be selected from the subspace pair alone exactly on simple nonzero principal spectra. If a nonzero principal value has multiplicity $m \geq 2$, no natural orthonormal principal frame is determined by the subspace pair alone.*

Proof. If the nonzero eigenvalues of K_+ are simple, their eigenspaces are complex lines. Choosing unit vectors in these lines is unique up to phase. Applying the polar partial isometry gives the matched principal lines in S_- . Thus principal frames are canonical up to independent phases.

If a nonzero eigenvalue has multiplicity $m \geq 2$, the corresponding eigenspace carries an internal $U(m)$ freedom. In the model block where $C = \sigma I$ on an m -dimensional principal block, every orthonormal basis is equally compatible with the same subspace-pair data. A natural frame rule depending only on the subspace pair would have to be fixed by all of $U(m)$, which is impossible for an ordered orthonormal basis when $m \geq 2$. $\square$

Corollary 5.6 (Finite-dimensional completeness). *For finite-dimensional subspace pairs, the endpoint dimensions and the principal-angle multiset classify the pair up to ambient unitary equivalence.*

Proof. This is the finite-dimensional CS decomposition. The pair splits into common, orthogonal, residual, and principal two-plane blocks. The dimensions and principal angles determine the block model, and any pair with the same data is unitarily equivalent to that model. $\square$

Research gap 5.7 (GAP-5A: theorem packaging). *The endpoint operator package is now mathematically stable, but the final paper should package it as one clean theorem with named corollaries: projector test, polar transport, frame criterion, biunitary finality, and CS completeness. This is mainly an exposition gap, not a proof gap.*

6. A_r VERSUS O_r

Two related subspace filtrations occur throughout the paper:

$$O_r = \text{span}\{P\partial^\alpha\psi : 1 \leq |\alpha| \leq r\},$$

and

$$A_r = \text{span}\{\psi, \partial^\alpha\psi : 1 \leq |\alpha| \leq r\}.$$

The first is horizontal and value-channel-free. The second keeps the state line and is often the cleaner ambient object.

Proposition 6.1 (Role split). *For each endpoint,*

$$A_r = \text{span}\{\psi\} \oplus O_r.$$

The orthogonal projector $P = I - |\psi\rangle\langle\psi|$ restricts to a surjection $A_r \rightarrow O_r$ with kernel $\text{span}\{\psi\}$. Thus O_r is canonically the quotient of A_r by the state line.

Proof. Each derivative decomposes into its horizontal projection plus a state-line component. This gives $A_r \subseteq \text{span}\{\psi\} + O_r$, while the reverse inclusion is immediate from the definition of O_r . Orthogonality of $\text{span}\{\psi\}$ and O_r follows from the horizontal projection. The kernel and quotient statement follow from the restriction of P . $\square$

There is no global winner between these two objects. O_r is the faithful horizontal object, closest to the value-channel-removed QGT pattern. A_r is the ambient extension and supports the cleanest transport-free endpoint theorem when all endpoints live in a common Hilbert space. Applications should state which role is being used.

Research gap 6.2 (GAP-6A: physical selection principle). *The current theorem is a role split, not a selection principle. A stronger physics-facing paper would identify settings where A_r or O_r is forced by an operational or dynamical criterion. For now, the paper keeps both and states the distinction explicitly.*

7. DYNAMICS AND OPERATIONAL READOUT

7.1. Exact unitary orbits and Krylov spaces. Let $\psi(t) = e^{-itH}\psi_0$. Then $\partial_t^k \psi(t) = (-iH)^k e^{-itH}\psi_0$, so

$$A_r(t) = e^{-itH} \text{span}\{\psi_0, H\psi_0, \dots, H^r \psi_0\}.$$

The horizontal space $O_r(t)$ is the value-channel-removed counterpart:

$$O_r(t) = e^{-itH} (\text{span}\{\psi_0, H\psi_0, \dots, H^r \psi_0\} \cap \psi_0^\perp).$$

Thus exact unitary dynamics turns the abstract osculating spaces into transported Krylov and compressed-Krylov spaces.

If the cyclic space of (H, ψ_0) has dimension m , then the spaces stabilize once $r \geq m - 1$. In a finite-cyclic case, the spectral measure of (H, ψ_0) gives an orthogonal-polynomial basis and a finite Jacobi matrix J . In that basis, the full ambient unitary package is represented by $e^{-i\Delta J}$. This is a specialization of the subspace theorem, not a general matrix theorem.

7.2. Projector speed and leakage. For a unitary subspace orbit $S(t) = e^{-itH}S(0)$, let Π_t be the orthogonal projector onto $S(t)$. Then

$$\dot{\Pi}_t = -i[H, \Pi_t]$$

and

$$\frac{1}{2} \|\dot{\Pi}_t\|_{HS}^2 = \frac{1}{2} \|[H, \Pi_t]\|_{HS}^2 = \|(I - \Pi_t)H\Pi_t\|_{HS}^2.$$

The Grassmannian speed is the Hilbert-Schmidt size of the Hamiltonian leakage block. At finite separation,

$$\Pi_t \Pi_{t+\Delta} \Pi_t|_{S(t)} = I - \Delta^2 L_t^* L_t + O(\Delta^3), \quad L_t = (I - \Pi_t)H\Pi_t.$$

7.3. Kato transport and curvature. Every smooth constant-rank projector path admits the Kato Hamiltonian

$$H_K(t) = i[\dot{\Pi}(t), \Pi(t)].$$

Its unitary propagator realizes the path:

$$\Pi(t) = U_K(t)\Pi(0)U_K(t)^*.$$

This gives a canonical pathwise model. It is not an endpoint-only comparison. In several parameters the Kato connection has curvature

$$R^K = \Pi(d\Pi \wedge d\Pi)\Pi.$$

Nonzero curvature produces holonomy and obstructs endpoint-independent Kato transport. In rank one this reduces to Berry curvature.

7.4. Reflection protocol. The endpoint operator package also has a coherent readout under an explicit access model.

Proposition 7.1 (Alternating reflections read principal angles). *Let $S_{\pm}$ be finite-dimensional endpoint subspaces with projectors $\Pi_{\pm}$. Suppose the reflections*

$$R_{\pm} = 2\Pi_{\pm} - I$$

can be implemented coherently. On each nontrivial principal two-plane of the pair, the unitary $W = R_-R_+$ has eigenphases $\pm 2\theta_i$, where θ_i is the corresponding principal angle.

Proof. On a principal two-plane choose an orthonormal basis e, f with

$$e \in S_-, \quad S_+ \cap \text{span}\{e, f\} = \text{span}\{\cos \theta e + \sin \theta f\}.$$

In this basis

$$R_- = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad R_+ = \begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}.$$

Thus

$$R_-R_+ = \begin{pmatrix} \cos 2\theta & \sin 2\theta \\ -\sin 2\theta & \cos 2\theta \end{pmatrix},$$

whose eigenvalues are $e^{\pm 2i\theta}$, up to the convention for rotation orientation. $\square$

Intersection sectors give phase +1, one-sided orthogonal residual sectors give phase -1, and ambient padding must be separated from the endpoint principal-angle data. Controlled phase estimation on W estimates the principal angles when coherent reflection access and adequate state preparation or sampling are available. This is a protocol for reading the existing operator package. It is not a new invariant and not a universal Hamiltonian model for arbitrary jet families.

Research gap 7.2 (GAP-7A: beyond reflection access). *The reflection protocol is natural in quantum algorithmic settings with coherent membership tests, code-space reflections, symmetry-sector reflections, or phase-kickback access. It may be too strong for arbitrary jet-derived subspaces. A stronger physics paper should find a concrete Hamiltonian, measurement, or response model where the same principal angles arise without assuming arbitrary projector reflections.*

8. BENCHMARK FAMILIES

The theory is not only a no-go theorem. The surviving subspace data can be richer than endpoint overlap.

8.1. Oscillator sanity check. For the harmonic-oscillator ground state

$$\psi_\omega(x) = \left(\frac{\omega}{\pi}\right)^{1/4} e^{-\omega x^2/2}, \quad \omega > 0,$$

the same-point first-jet block in the ω -coordinate is

$$M(\omega) = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{8\omega^2} \end{pmatrix}.$$

The endpoint overlap is

$$I(\omega_-, \omega_+) = \frac{\sqrt{2}(\omega_- \omega_+)^{1/4}}{\sqrt{\omega_- + \omega_+}}.$$

The raw first-jet blocks change under the squeeze parameter $r = \frac{1}{2} \log \omega$, but the bilaterally whitened first-jet block stays unchanged in this benchmark. Its singular values collapse to a function of overlap alone, so the oscillator is a sanity check rather than a richer-than-overlap example.

8.2. Qutrit polynomial curve. Consider

$$\psi(t) = \frac{(1, t, t^2)^T}{\sqrt{1 + t^2 + t^4}}.$$

Writing

$$D(t) = 1 + t^2 + t^4, \quad J(t) = 1 + 4t^2 + t^4,$$

the identity-transport representative of the first-horizontal whitened block is

$$\Omega(u, v) = \begin{pmatrix} \frac{1+uv+u^2v^2}{\sqrt{D(u)D(v)}} & \frac{(u-v)(1+2v^2+2uv+uv^3)}{\sqrt{D(u)D(v)J(v)}} \\ \frac{(v-u)(1+2u^2+2uv+u^3v)}{\sqrt{D(u)D(v)J(u)}} & \frac{1-u^4-v^4+5uv+4u^3v+4uv^3+5u^3v^3+u^4v^4}{\sqrt{D(u)D(v)J(u)J(v)}} \end{pmatrix}.$$

Its singular values are 1 and

$$\frac{|1 + 4uv + u^2v^2|}{\sqrt{(1 + 4u^2 + u^4)(1 + 4v^2 + v^4)}}.$$

The pairs

$$(u, v) = \left(-5, \frac{6}{5}\right), \quad (u', v') = \left(-\frac{5}{2}, -\frac{3}{5}\right)$$

have the same overlap

$$I(u, v) = I(u', v') = \frac{25\sqrt{39}}{273},$$

but different second singular values. Thus the first-order endpoint geometry sees more than ordinary overlap in this benchmark.

8.3. Quartit cubic and O_2 . For

$$\psi(t) = \frac{(1, t, t^2, t^3)^T}{\sqrt{1 + t^2 + t^4 + t^6}},$$

the ambient second osculating space is a varying 3-plane in $\mathbb{R}^4$, with normal

$$n(t) = (-t^3, 3t^2, -3t, 1)^T.$$

The surviving ambient singular values are

$$1, \quad 1, \quad \frac{|1 + 9uv + 9u^2v^2 + u^3v^3|}{\sqrt{(1 + 9u^2 + 9u^4 + u^6)(1 + 9v^2 + 9v^4 + v^6)}}.$$

There are endpoint pairs with the same overlap and different final singular value.

The value-channel-free side is more informative. For

$$O_2(t) = \text{span}\{j_1(t), j_2(t)\},$$

the principal-angle cosines of $O_2(u)$ and $O_2(v)$ are the singular values of

$$M_{O_2}(u, v) = \begin{pmatrix} \frac{1+uv+u^2v^2+u^3v^3}{\sqrt{D_4(u)D_4(v)}} & \frac{(u-v)^3}{\sqrt{D_4(u)N_4(v)}} \\ -\frac{(u-v)^3}{\sqrt{N_4(u)D_4(v)}} & \frac{1+9uv+9u^2v^2+u^3v^3}{\sqrt{N_4(u)N_4(v)}} \end{pmatrix},$$

where

$$D_4(t) = 1 + t^2 + t^4 + t^6, \quad N_4(t) = 1 + 9t^2 + 9t^4 + t^6.$$

On this benchmark O_2 is strictly finer than A_2 : the ambient invariant keeps only the product of the two O_2 singular values, while O_2 keeps both.

8.4. Veronese family. Let

$$v_n(t) = (1, t, \dots, t^n).$$

At top order $r = n - 1$, the ambient osculating subspace is a codimension-one hyperplane. Its normal is the coefficient vector of $(x - t)^n$. Therefore the single nontrivial principal-angle cosine is the normalized binomial-square polynomial in uv . For every $n \geq 2$, this quantity is generically not a function of endpoint overlap alone.

On the value-channel-free side, O_{n-1} has a codimension-two complement description. The corresponding 2×2 complement matrix recovers the qutrit O_1 and quartit O_2 cases and is generically richer than overlap. Lower-order

$r < n - 1$ complements are known structurally, but closed spectral formulas are not yet part of the paper.

8.5. Spherical twist family. The phenomenon is not confined to polynomial curves. For

$$\psi_k(t) = (\cos t \cos kt, \cos t \sin kt, \sin t), \quad k \geq 2,$$

there are exact equal-overlap endpoint pairs with different A_1 and O_1 principal-angle data. For $k = 2$, the pairs $(0, \pi/4)$ and $(0, \pi/2)$ both have overlap 0, but the corresponding A_1 and O_1 angle data differ.

8.6. Finite-jet genericity for O_1 . The examples above have a local generic mechanism. For normalized real C^2 curves in finite dimension at least 3, fix two distinct parameter values with nonzero endpoint velocities. Define

$$I(u, v) = \langle \psi(u), \psi(v) \rangle, \quad T(t) = \frac{\psi'(t)}{\|\psi'(t)\|}, \quad C(u, v) = \langle T(u), T(v) \rangle.$$

On a branch where $C \neq 0$, the first value-channel-free line-angle observable is locally independent of overlap whenever

$$\det \frac{\partial(I, C)}{\partial(u, v)} \neq 0.$$

After clearing the nonzero velocity denominators, this determinant is a polynomial in the endpoint 2-jets and is not identically zero on the admissible jet chart. Thus nonvanishing is an open dense finite-jet condition. Consequently, for smooth curve families whose endpoint 2-jet map is submersive onto such a chart, and for real-analytic families where the pulled back determinant is not identically zero, there is an open dense locus where the O_1 line-angle data is not locally determined by overlap alone.

Research gap 8.1 (GAP-8A: higher-order genericity). *The finite-jet theorem is first-order and value-channel-free. A stronger paper would prove generic richer-than-overlap behavior for higher O_r and A_r principal-angle spectra, and would identify the sharp dimension and regularity hypotheses.*

Research gap 8.2 (GAP-8B: lower-order Veronese spectra). *The lower-order Veronese complement spaces have structural descriptions, but closed or conceptual formulas for their full principal-angle spectra remain to be written. This is a concrete algebraic target for the next research cycle.*

9. LIMITS AND RESEARCH PROGRAM

The theory has a clean core and clear boundaries.

- (1) It does not produce a canonical full matrix from higher jets. It explains why such a matrix is absent without extra frame-selecting structure.
- (2) It does not give endpoint-independent Kato transport in nonzero curvature. Kato transport is pathwise.

- (3) It does not choose principal frames in repeated-spectrum sectors.
- (4) It does not yet supply a physics case study where higher osculating subspace angles reveal a new material, algorithmic, or dynamical effect beyond known QGT/fidelity diagnostics.

The next research program is therefore concrete.

Research gap 9.1 (GAP-10A: physics case study). *Find a concrete quantum system where the O_r/A_r principal-angle package reveals information not visible to ordinary overlap, fidelity susceptibility, or the first-order quantum geometric tensor. Candidate settings include Bloch state families, flat-band geometry, topological phase boundaries, variational ansatz manifolds, and Krylov dynamics.*

Research gap 9.2 (GAP-10B: mixed states and degenerate bands). *Extend, or sharply fail to extend, the osculating-subspace package to density matrices, purification bundles, or degenerate Bloch bands. This must be compared with non-Abelian quantum geometry and Wilczek-Zee transport before any novelty claim is made.*

Research gap 9.3 (GAP-10C: numerical invariant library). *Build a stable numerical pipeline that takes a sampled state family, estimates the O_r and A_r endpoint subspaces, and computes principal-angle spectra with conditioning diagnostics. This would turn the theory into a usable diagnostic tool.*

10. CONCLUSION

The attempted higher-order matrix generalization of the quantum geometric tensor fails for principled reasons. Higher projected jets mix by non-unitary triangular laws under phase and coordinate changes, so raw matrix entries are not canonical. The failure identifies the correct invariant object: the osculating subspace filtration and the endpoint principal-angle/operator package.

This object is mathematically rigid. Its frame matrices form a biunitary orbit, its singular values are principal-angle cosines, its polar factor is canonical on the nonzero-overlap sector, and its finite-dimensional classification is the CS decomposition. It is also useful: explicit benchmarks show it can distinguish endpoint pairs with the same ordinary overlap, and coherent alternating reflections read it under standard access assumptions.

The remaining work is not to force a noncanonical matrix into the theory. It is to use the canonical subspace/operator package in serious physics examples.

PROVENANCE NOTE

This draft is assembled from the working note, the paper skeleton, and the reports in the active quantum team directory. The most important deposits are the reports on the first-order QGT bridge, the higher-jet no-go,

covariant jets, the endpoint operator package, biunitary matrix finality, the reflection protocol, and nonbenchmark genericity.